Arsh Dhawan

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BS-MS
LinkedIn
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EDUCATION

- **BS-MS**
  Physical Sciences — 2023 - Present
  IISER Bhopal, India
- **Higher Secondary / +2**
  Adarsh Jain, Delhi — 2023
- **Secondary**
  Scholars Rosary, Rohtak — 2021

About

I am a third-year Physics Student interested in pursuing Theoretical Physics, espescially interested in pursuing QFT and Mathematical Physics in future. I find both physics and mathematics fascinating espescially topics like GR, QFT and Group Theory, and try to approach them by their most basic postulates My experience under Prof. Ganguli has made me more inclined to understand the subject by focusing on essence of the subject, which translates better in Physics and Mathematics than in any other field. Moreover the opportunity under Dr. Naga Deepthi made me familiar with current advancements in Neutrino Physics while Dr. Rahul introduced me to Particle Physics and BSM Phenomenology which I briefly worked upon during Dec'25-Mar'26.
I am also familiar with a few specialized coding languages like GLoBES and SNoWGLoBES as well as proficient in Mathematica and MATLAB. Though, I am familiar I wish not to work in coding heavy projects and am more into mathematics/theory and phenomenology as well as model building. I am willing to do some coding for realization of these things and will most probably work alongside Arnav Kapoor for the same.

Experiences

- **Summer Internship 2026; Neutrino NSI Interactions and Sterile Neutrinos** — Dr. Deepthi K. N.
  Under Dr.Deepthi, I continued my previous project on Scalar NSI resulting in pre-print, ' '. Other than that I worked on understanding Sterile Neutrinos, one of the contenders of Dark Matter which shall have impact on my project(s) with Dr. Rahul Srivastava in near future.
- **Project: Discrete Symmetries and Generating Neutrino Masses (SeeSaw Mechanism) by Discrete Symmetry Extensions of Standard Model** — Dr. Arnab Chaudhury and Dr. Mitesh Behera
  Under Dr. Arnab Chaudhury and co-supervisor Dr. Behera, I understood the Discrete Symmetry Groups and their importance in Particle Physics.
  This exercise was followed by extended Standard Model using Discrete Symmetry group, <Name> and then trying to generate mass of neutrinos in this new extension of standard model by Type-I,II,III See-Saw models
- **DM** — Dr. Rahul Srivastava
  My last introductory project with Dr. Rahul prompted us to pursue another project where I attempted to understand...(To be updated)
- **Non-Hermitian Dynamics in 2-flavour Neutrino Oscillations(Summer 2026)** — Dr. Poonam Mehta
  Under Dr. Poonam Mehta I am woking on a project which as planned would attempt in studying Non Hermitian Physics starting from OQS, and better understanding Entanglement and Bell Theorem.
  Following this we shall attempt to use G-metric and Density Matrix approach in Non-Hermitian Physics for 2-flavour neutrino-oscillations. This project is heavily influenced by Dr. Poonam's recent paper on
- **Project: Dark Matter and Its Contenders(2025-26)** — Dr. Rahul Srivastava

  During winter(Dec-Mar'26), I took up a project under Dr. Rahul Srivastava, Associate Professor, IISER Bhopal where I approached

Dark Matter Phenomenology using Particle Physics(BSM). For this I referred to Dr. Gianfranco Bertone white paper and book on the same alongside other resources. Other than this I also worked on limiting cases of neutrino physics under constraints imposed by solar, atmospheric and Supernova neutrinos.

- **ICTP PWF Riemannian and Complex Geometry (2025-26)** ICTP

  ICTP offers a select few students, courses in a few topics every year. During Sep'25-Feb'26, I was part of one such course focused upon "Riemannian and Complex Geometry" which introduced me to topology, differential, riemannian and complex geometry along with some concepts/practical applications in General Relativity.

- **Teaching Assistant, Quantum Physics (2025)** Dr. Dr.Nirmal Ganguli

  Under Prof. Ganguli, I acted as TA for Quantum Physics course, helping other students understand the topics better by solving their doubts and sometimes, teaching a few during the Tutorials.

- **Summer Internship 2025; Supernovae Neutrinos and Neutrino Oscillations** Dr. N. K. Deepthi

  Under Dr. Naga Deepthi, I worked in the field of Neutrino Oscillations covering the MSW effect in matter. Later, I focused on the study of supernova and the neutrino emission during such astrophysical events. I was also introduced to the fundamental concepts of Open Quantum Systems as well as basic QFT and Particle Physics, required for the project. I also delved into Computational aspect, making use of GLoBES to understand neutrino phenomenology better. Later following the collaboration, I studied Amol Dighe's paper on 'Identifying FLux from Supernova Explosions' as well as Kuo and Mikheyev's paper on Resonance in Neutrino Oscillations.

- **Reading Project; Quantum Mechanics** Dr. Arnab Rudra

  Under Dr. Rudra, in an extended program, he encouraged learning quantum mechanics by a problem solving approach, making us familiar with basics of QM as well as dirac notation.

- **Summer School; General Relativity and Cosmology** Dr. Shibesh Kumar Jas, Director, PICS

  Under Dr. Shibesh, I got familiar with General Relativity and Black Hole Thermodynamics, using computational Tools like Mathematica throughout this project.

Working Paper

- **NSI Scalar Interaction in Neutrino Physics**
  I am currently working on a paper with Chinmay Bera(Ph.D) and Dr. Deepthi N. K. on finding mixing matrices as well as effective masses when considered Vacuum, matter and NSI Scalar effects on Neutrino Oscillation simultaneously. It is a continuation of a paper by the latter 2 with Prof. R. Mohanta. You can find the methodology and updates here.

TECHNICAL SKILLS

- **Linux**
- **C**
- **MATLAB**
- **Mathematica**
- **LaTeX**
- **GLoBES**
  SNowGLoBES for Neutrino Physics

Relevant Courses

- **Mechanics**
  B+

- **Quantum Physics**
  A
- **Calculus**
  B
- **Groups and Symmetries**
  B
- **Quantum Mechanics-I**
  B
- **Standard Model of Particle Physics**
  A [Audited]

Knowledge from other Sources

- **Relativistic QM**
- **Basic QFT**
- **Open Quantum Systems**

Extra-Curricular Activities

- **Playwright** 2023-24
  AARAMBH [IISER-B Drama Club]
- **Head Writer** 2024-2025
  AARAMBH[IISER-B Drama Club]
- **Content writer** 2023-24
  Singularity Tech Fest [IISER-B]
- **Content writer** 2023-24
  Mahuaa Fest [IISER-B]

Volunteer

- **Literary Fest, Arts and Letters** 2024
- **Integration Bee** 2024
  Acted as event organizer of "Integration Bee" for IISER-B Math Fest, Continuum.